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Research Paper

Toward Waste-Minimising Biorefineries: Integrating Fermentation and Hydrothermal Fulvification of Grass

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ABSTRACT

Biochemistry-based biorefineries frequently valorize agricultural residues but rarely achieve full carbon utilization, leaving substantial downstream solids and liquids underexploited. This study presents an integrated fermentation–hydrothermal fulvification (HTF) strategy that upgrades grass biomass into lactic acid and artificial humic substances. Acid pretreatment generated 45.1 g/L total sugars and up to 36 g/L lactic acid, while base-pretreated substrates yielded 37.1 g/L of total sugars and 29.7 g/L lactic acid, with lower inhibitor levels. Subsequent HTF under strongly alkaline conditions converted lignin-rich fermentation residues into hydrochars with yields ranging from 21.2 to 45.4 wt.% and produced artificial humic acids up to 10.8 wt.% with elevated carbon contents of 74.5 wt.%. A pronounced decrease in O/C ratio indicates progressive dehydration and condensation reactions. Process liquids reached up to 36.2 g/L total organic carbon and contained organic acids, including a chromatographic peak assigned to lactic acid up to 9.9 g/L, likely formed in situ under alkaline HTF conditions, alongside phenolic compounds up to 1121 mg/L. Spectroscopic and colloidal analyses confirmed partitioning into humic-rich solids and fulvic-like dissolved organics. Humic acid yields from post-fermentation residues remained comparable to standalone HTF of raw grass, indicating that upstream lactic acid production does not compromise downstream humification. By converting the remaining solid residues through HTF into artificial humic substances that do not require further separation, the integrated system supports a waste-minimising biorefinery approach, in which all major fractions are functionally utilised as products while reducing biological risks and potential greenhouse gas emissions associated with untreated residues.

Abbreviation

A-FA	Artificial Fulvic Acid	HMF	5-Hydroxymethylfurfural
A-HA	Artificial Humic Acid	LA	Lactic Acid
EEM	Excitation Emission Matrix	PLA	Poly-lactic Acid
FA	Fulvic Acid	SY	Solid Yield
HA	Humic Acid	TC	Total Carbon
HAY	Humic Acid Yield	TOC	Total Organic Carbon
HTC	Hydrothermal Carbonization	TS	Total Solids
HTF	Hydrothermal Fulvification		

1. Introduction

Grass represents one of the most plentiful renewable biomass resources, as grasslands occupy nearly 30-40% of the global land surface (Blair et al., 2014). Its chemical composition, dominated by cellulose, hemicellulose, and lignin (Sun and Cheng, 2002), makes grass a promising feedstock for diverse conversion routes. While grasses have traditionally served as animal feed, their chemical composition also enables conversion into diverse products through thermo-chemical and biochemical (Tursi, 2019), and thermochemical processes (Ghobadian

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et al., 2026), as well as through purely physical processing, where mechanical pressing yields a fibre-rich solid fraction suitable for use in materials, solid biofuels, or further biorefinery applications (Moloeznik Paniagua et al., 2023). Beyond these technical possibilities, grasses represent an underexplored resource for innovative bio-based products, such as paper, plastics, and other materials. Yet, commercialization remains limited due to technological, financial, and consumer awareness barriers (Orozco et al., 2021). The transition toward sustainable and integrated biomass conversion has intensified interest in waste-minimising and cascading valorisation approaches that seek to maximise the utilisation of all biomass fractions (Arevalo-Gallegos et al., 2017; Marzban et al., 2025), while aligning with socio-economic and policy frameworks to support emerging grass-based businesses (Orozco and Grundmann, 2022). Recent advances in integrated biorefinery concepts further emphasize the cascading utilization of biomass fractions to maximize carbon efficiency and product diversity (Chong et al., 2024).

However, despite significant advances in biomass conversion technologies, most biorefinery concepts still fail to achieve full carbon utilization, as substantial fractions of lignocellulosic feedstocks remain under-valorized after primary processing steps. Addressing this limitation requires integrated strategies that sequentially upgrade both carbohydrate- and lignin-rich fractions within a unified framework (Marzban et al., 2025). In particular, downstream solid and liquid streams are often treated as secondary residues rather than as integral components of the overall product system, limiting overall resource efficiency.

Fermentation is a central upstream process in the biorefinery concept, enabling the selective conversion of fermentable sugars into industrially relevant biochemicals (Ewing et al., 2022). It is capable of generating a variety of organic acids, among which succinic acid and malonic acid are well-recognized, while lactic acid has gained particular prominence as the most significant and widely utilized product. Lactic acid (LA) is a valuable organic compound widely used in the food, pharmaceutical, and cosmetic sectors (Wang et al., 2015). Optically pure lactic acid, particularly the L-isomer of LA, serves as a crucial intermediate for synthesizing poly-lactic acid (PLA), a biodegradable and bio-based polymer widely used in the production of environmentally friendly materials and packaging applications (Swetha et al., 2023) . The conventional routes of lactic acid production make extensive use of starch-rich feedstocks and purified sugars, and this dependence on food-grade substrates not only limits sustainability but also accounts for a considerable share of the overall production costs (Herrmann et al., 2024). Agricultural residues such as grass offer an attractive non-food alternative, as they can be hydrolyzed into mixed sugars and efficiently fermented into LA by microorganisms such as *Bacillus coagulans* (Ma et al., 2016).

Nevertheless, lignocellulosic fermentation primarily targets the carbohydrate fraction. As a result, lignin-rich solid residues remain and retain a considerable portion of the original biomass carbon. These residues include digestate from anaerobic digestion, lignin-rich by-products from lignocellulosic ethanol production (Santos et al., 2015), and stillage from bioethanol processing (Scarborough et al., 2018). They are often applied as soil amendments or fertilizers. However, unconverted biomass components may lead to undesirable greenhouse gas emissions (Marzban et al., 2025). Importantly, these residues are rich in lignin and contain aromatic structures and phenolic functional groups that closely resemble those found in natural humic acids (Marzban et al., 2024; Sarlaki et al., 2026). The aromatic backbone and phenolic functionalities of these residues provide chemically suitable precursors for condensation and polymerization reactions under alkaline hydrothermal conditions, making them promising substrates for humification-oriented upgrading (Sarlaki et al., 2026).

Among biomass valorization strategies, hydrothermal fulvification (HTF) has recently emerged as a modified alkaline approach to conventional hydrothermal carbonization (HTC) (Marzban, 2024). While HTC efficiently converts wet biomass into hydrochar, liquid, and gaseous products (Libra et al., 2011), its liquid phase is often considered a problematic by-product due to high total organic carbon loads and poorly characterized, potentially phytotoxic compounds (Petrović et al., 2024). In contrast, HTF operates under strongly alkaline conditions (exceeding two molar equivalents of KOH per carbohydrate unit), steering biomass conversion toward artificial humic and fulvic acids rather than aromatic condensates (Marzban et al., 2024; Tkachenko et al., 2023). Under these alkaline conditions, lignin fragmentation and retro-aldol reactions of residual carbohydrates generate reactive carbonyl and phenolic intermediates that undergo secondary condensation into humic-like macromolecular networks (Ischia et al., 2026). The resulting artificial humic substances exhibit functional properties comparable to naturally occurring humic and fulvic acids (Marzban et al., 2024), ubiquitous constituents of soils and aquatic systems (De Melo et al., 2016), which are known for their roles as metal chelators, redox mediators, soil conditioners, and plant growth promoters (Sarlaki et al., 2026), as well as for their influence on the soil microbiome (Hoefle et al., 2024). Unlike natural humic substances, which are typically derived from non-renewable sources such as lignite and peat (Zherebtsov et al., 2019), HTF enables their production from renewable biomass, and its dilute liquid fractions have additionally been reported to promote seed germination (Ghobadian et al., 2026). Following the introduction of the main concept of artificial humic substances (Yang et al., 2019; Yang and Antonietti, 2020), these materials have been increasingly recognized for their potential in soil and environmental applications and were recently listed by IUPAC among the top ten emerging technologies (Gomollón-Bel, 2021).

Within this conceptual framework, artificial humic substances are not isolated fractions but represent a continuum of soluble and insoluble products, which can be functionally integrated rather than separated, thereby supporting a waste-minimising system design.

This work, to the best of our knowledge, represents the first investigation into the multistage valorization of grass biomass, in which lactic acid fermentation is directly coupled with hydrothermal fulvification to generate artificial humic substances. After fermentation, the remaining solid residues, still containing structural carbohydrates, lignin fragments, and mineral components, are typically underutilized or subjected to direct land application, where microbial degradation may result in greenhouse gas emissions. In this study, these residues were instead subjected to a controlled hydrothermal fulvification process under strongly alkaline conditions. This thermochemical step enables the transformation of residual lignocellulosic fractions into artificial humic and fulvic acids, thereby integrating fermentation and hydrothermal processing into a continuous valorization pathway. By linking upstream biochemical conversion with downstream thermochemical upgrading, the proposed approach contributes to a waste-minimising biorefinery concept, where all major process streams are valorised within a unified system rather than treated as residual waste. The significance of this approach lies in linking biochemical and thermochemical conversion pathways within a single framework, enabling simultaneous valorisation of carbohydrate- and lignin-rich fractions rather than treating downstream residues as waste.

2. Materials and methods

2.1. Raw materials and experimental process

Grass was collected from periodically flooded polder land in the LOV National Park near Berlin (53°02′32.6″N 14°17′50.0″E) and subsequently processed in the field through cutting, drying, turning, rowing, and baling (Heinrich et al., 2023). For fermentation experiments, the thermophilic microbial isolates A162 and A166 were employed, which were identified as *Heyndrickxia coagulans* (formerly *Bacillus coagulans*) by the Leibniz Institute DSMZ-German Collection of Microorganisms and Cell Cultures (Braunschweig, Germany) and are currently maintained at the

Leibniz Institute for Agricultural Engineering and Bioeconomy (Potsdam, Germany). The overall experimental process designed in this study is presented in Fig. 1. After the preparation of raw material, the workflow consisted of two main stages: fermentation and hydrothermal fulvification. Fermentation was carried out to convert the biomass into lactic acid, while the subsequent hydrothermal fulvification step transformed the material into humic substances.

2.2. Hydrolysis process

Grass biomass was subjected to chemical-thermal pretreatments under three conditions: acid, base, and base-washed. Pretreatments were conducted in Schott bottles using an autoclave (Tuttnauer 3150ELV), with 1% (w/w) H_2SO_4 for acid treatment and KOH for base treatment. In the base-washed scenario, the pretreated material was centrifuged (Sigma 4-16KS, 4800 rpm, 15 min) prior to enzymatic hydrolysis.

The hydrolysis process consisted of liquefaction followed by saccharification, performed in 2 L BIOSTAT bioreactors (Sartorius AG, Germany) with a working volume of 1 L. Grass suspensions (10% w/v) were processed under unsterile conditions, and the pH was adjusted to 6.0 using either 25% (w/w) KOH or 50% (w/w) H_2SO_4.

Cellic CTec 2 was supplemented in the hydrolysis vessel at a dosage of 30% v/w per g of cellulose, and enzymatic conversion proceeded for 48 h. Upon completion, the resulting mixture was manually filtered through a 150 μm pore-size filter bag, followed by centrifugation at 4800 rpm for 15 min to yield the hydrolysate used in fermentation experiments.

2.3. Fermentation process

Inoculum cultures were prepared in MRS broth supplemented with 0.67 g EVERZIT Dol (0.5-2.5 mm) and one MRS agar slant, followed by incubation at 40 °C and 100 rpm for 16 h to obtain active seed inocula for fermentation. Fermentation was performed in duplicate using an Eloferm multifermentation system (EloSystems GbR, Germany) with a working volume of 500 mL. The process was conducted at 52 °C under continuous stirring at 400 rpm, using grass hydrolysate as the substrate. The pH was maintained at 6.0 throughout fermentation.

2.4. Hydrothermal fulvification

The hydrothermal fulvification (HTF) experiments were conducted using the residual solids obtained after lactic acid extraction from grass. To ensure that the reaction proceeded via hydrothermal fulvification (HTF) rather than hydrothermal carbonization (HTC), the reaction medium was adjusted to contain a molar excess of alkali, with the KOH loading exceeding two molar equivalents. This alkali requirement was calculated based on the structural carbohydrate fraction of the biomass, which was determined using the detergent fiber system. Neutral detergent fiber (NDF) represents the total cell wall fraction, comprising hemicellulose, cellulose, and lignin, while acid detergent fiber (ADF) consists of cellulose and lignin only, and acid detergent lignin (ADL) quantifies lignin. Accordingly, hemicellulose was calculated as NDF minus ADF, cellulose as ADF minus ADL, and lignin was equated to ADL.

Using this approach, clear differences in fiber composition were observed between grass and the fermented solid residues obtained after acid (AWS) and base (BWS) treatments with post-washing, as well as the unwashed residue after base treatment (BNWS) (Table 1). Grass exhibited substantially higher structural carbohydrate contents than the treated residues, a distinction that is relevant for evaluating feedstock reactivity and alkali demand under HTF conditions.

For each experiment, 677.9 g of feedstock slurry (60.8 g grass, 16.9 g KOH and 600.2 g distilled water, corresponding to a biomass-to-water ratio of 0.1 (w/w), was loaded into a 1 L high-pressure reactor (Parr Instrument, Series 4520, Moline, Illinois, USA), sealed, and heated to 230 °C at a rate of 2 K/min, stirred at 200rpm, with a residence time of 4 h.

Three experimental runs were performed: direct use of grass in hydrothermal fulvification (GF), grass subjected to acid pretreatment for

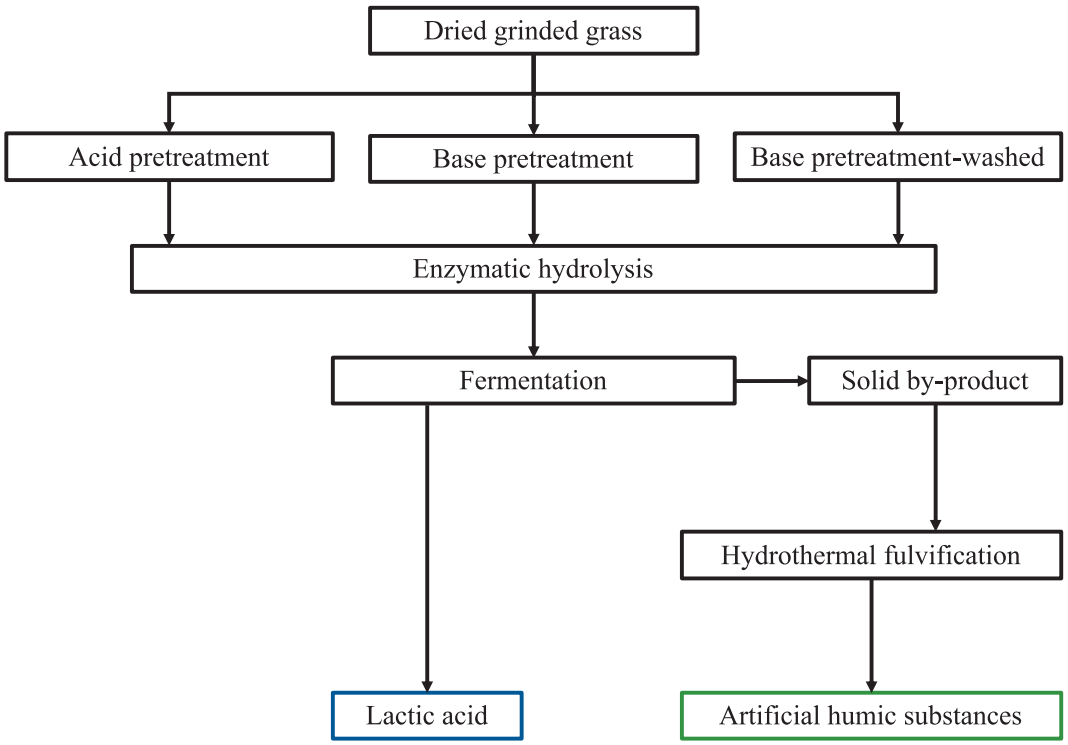


Fig. 1. Process flow for converting grass to lactic acid and humic substances (humin/hydrochar and liquid humic substances inlcuding humic and fulvic acids) by utilization of fermentation and hydrothermal fulvification.

Table 1
Detergent fiber fractions and calculated structural carbohydrates of grass and fermented solid residues after acid (AWS), base (BWS), and unwashed base (BNWS) treatments.

Sample	NDF (%)	ADF (%)	ADL (%)	Hemicellulose (%)	Cellulose (%)	Lignin (%)
Grass	64.8 ±2.2	36.4 ±1.2	5.6 ±0.2	28.4	30.7	5.6
AWS	16.0 ±0.3	13.4 ±0.3	6.2 ±0.2	2.6	7.2	6.2
BWS	11.7 ±0.4	9.6 ±0.3	2.9 ±0.1	2.1	6.7	2.9
BNWS	25.9 ±0.5	21.1 ±0.5	8.2 ±0.5	4.8	12.9	8.2

fermentation (LAF), and base pretreatment (LBF). In these three fermentation-derived samples, an additional washing step was applied prior to HTF to remove residual impurities. Furthermore, two additional base-pretreated samples without washing were tested, denoted LBF-NW-0.1 and LBF-NW-0.2, corresponding to 0.1 and 0.2 biomass-to-water ratios, respectively. Potassium hydroxide was added at four equivalents relative to the total carbohydrate content (cellulose + hemicellulose) to ensure effective hydrolysis. For example, the KOH-treated feedstock (biomass-to-water ratio of 0.2) comprised 483.6 g of fresh matter (total solids: 20.7%), 111.5 g of water, and 24.2 g of KOH, corresponding to four equivalents relative to the cellulose and hemicellulose content.

The rationale for the designation hydrothermal fulvification (HTF) is detailed in our previous works (Marzban et al., 2024; Tkachenko et al., 2023), where this concept was first introduced and clearly distinguished from hydrothermal humification (HTH), which employs 1-2 equivalents of KOH relative to carbohydrates, and hydrothermal carbonization (HTC), which operates with 0-1 equivalent.

Following HTF, the solid-liquid slurry was separated by filtration using flat filter paper (ROTH Type 113A-110, 5-8 μm). The recovered solid fraction was stored in vacuum-sealed plastic bags, whereas the liquid phase was collected in closed containers, preserved at −18 °C, and analyzed within 24 h.

The solid yield (SY) was determined as the proportion of input dry mass retained in the carbonized solid fraction, calculated on a total dry mass basis:

$$SolidYield\,(\%) = \frac{Massofproducedhydrochar}{Initialmassofbiomass} \times 100 \quad (1)$$

The extraction of artificial humic acids (A-HAs) from the liquid fraction was performed in duplicate following ISO 19822 guidelines with slight modifications. For each run, 10 g of process liquid was transferred into a 50 mL centrifuge tube and acidified with 32% HCl to adjust the pH to approximately 2. The acidified suspension was allowed to stand for 1 h and then centrifuged at 5000 rpm for 10 min to separate the solid and liquid phases. The resulting solid precipitate, corresponding to artificial humic acids (A-HA), was collected. The supernatant liquid phase was labeled and characterized as artificial fulvic acids (A-FA). The A-HA precipitate was subsequently dried in an oven at 60 °C for 48 h, weighed, and subjected to further analytical characterization. The yield of humic acids was expressed as the proportion of input dry mass recovered as A-HA, also calculated on a total dry mass basis, determined as the ratio of the extracted product in dry weight (M_{HA}) to the initial dry feedstock weight (M_b):

$$HAY\,(\%) = \frac{M_{HA}}{M_b} \times 100 \quad (2)$$

2.5. Characterizations and analytical methods

Cold-water extracts of the substrate were analyzed to quantify glucose, fructose, sucrose, and lactic acid (LA). The determination of sugars, LA, and formic acid was performed using high-performance liquid chromatography (HPLC; Dionex, Sunnyvale, CA, USA) equipped with a Eurokat H column (Knauer, Berlin, Germany) and a refractive index detector. Fiber composition was determined using the ANKOM2000 Fiber Analyzer (Ankom Technology Corp., Macedon, NY, USA) with filter bag technology. Total carbon (TC) and total organic carbon (TOC) in the HTF solid samples were quantified using a Shimadzu TOC-5050A analyzer. TOC values were determined by subtracting the measured total inorganic carbon (TIC) from the total carbon (TC). Elemental composition (C, H, N, and S) of the samples was determined using a Vario ELIII elemental analyzer (Elementar Analysensysteme, Hanau, Germany) with sulfonic acid as the reference standard. Samples were pre-dried at 60 °C prior to analysis. Oxygen content was calculated indirectly according to the equation: O% = 100 – (C% + H% + N% + S% + Ash%) (Wang et al., 2017). Aromatic compounds were analyzed using a Thermo Fisher ICS-3000 HPLC system equipped with a UV detector, whereas the inorganic content was determined by ICP-OES (Thermo Scientific iCAP6300 Duo) coupled with a CETAC ASX-520 autosampler. Sugars and organic acids were quantified using a Dionex Ultimate 3000 UPLC system fitted with a KNAUER Eurokat H column (300 mm × 8 mm, 10 μm).

Humic substances, including humic and fulvic acids, were characterized using UV-Vis spectroscopy, fluorescence spectroscopy, and particle size analysis. UV-Vis absorbance spectra were recorded with a Shimadzu UV-2600 spectrophotometer over the wavelength range of 200 to 800 nm. Excitation-emission matrix (EEM) fluorescence spectra were obtained with a Perkin-Elmer LS50B spectrophotometer, using excitation wavelengths from 250 to 500 nm at 10 nm intervals and emission wavelengths from 300 to 700 nm, with slit widths of 10 nm and a scanning speed of 1200 nm/min. Particle size distribution and zeta potential of the process liquids were determined with a Zetasizer Nano Ultra (Malvern Panalytical) equipped with backscattering detection.

3. Results and discussion

3.1. 9. Hydrolysis performance under different pretreatments

Enzymatic hydrolysis of pretreated grass biomass was evaluated using three different pretreatment approaches: acid pretreatment with H_2SO_4, base pretreatment with KOH, and base pretreatment combined with a washing step. **Fig. SI1** illustrates that acid pretreatment resulted in the highest total sugar release within the first 5 hours, after which the concentration stabilized (45.1 g/L), suggesting that most of the accessible cellulose and hemicellulose were hydrolyzed during the early stage. In contrast, base and base-washed pretreatments produced slightly lower final sugar concentrations of 36.7 and 37.4 g/L, respectively, yet exhibited comparable hydrolysis kinetics, reaching a plateau within approximately 10 h. The sharp initial increase observed across all treatments indicates that the pretreatments effectively disrupted the lignocellulosic structure, thereby improving enzyme accessibility.

To provide an indicative assessment of pretreatment efficiency, sugar recovery was estimated based on the initial structural carbohydrate content of the raw grass. Considering a total carbohydrate fraction (cellulose + hemicellulose) of approximately 59 wt.% and a biomass loading of 10% (w/v), the theoretical maximum sugar concentration corresponds to around 59 g/L. Based on the experimentally measured sugar concentrations, the acid pretreatment achieved a recovery of approximately 76%, while base and base-washed pretreatments resulted in recoveries of approximately 62-64%. These results confirm that acid pretreatment enhances carbohydrate solubilization, whereas alkaline pretreatment provides slightly lower sugar recovery but with reduced inhibitor formation.

Analysis of monosaccharide composition (Fig. 2-a) revealed glucose as the dominant hydrolysis product (19.3 to 24.2 g/L) across all pretreatments, indicating effective cellulose deconstruction. The acid-pretreated hydrolysate uniquely exhibited pronounced xylose release

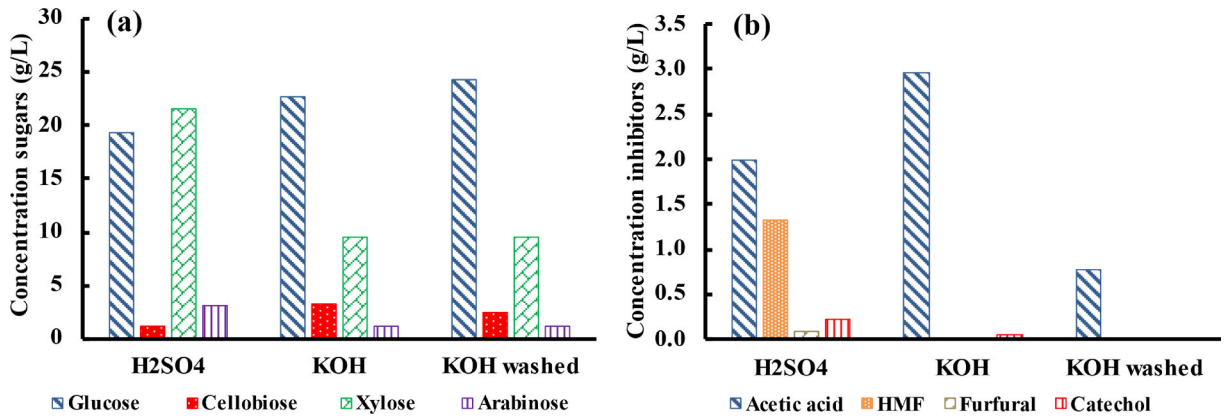


Fig. 2. Concentrations of a) individual sugars and b) fermentation inhibitors in hydrolysates derived from different pretreatments, after 48 hours enzymatic hydrolysis.

(21.5 g/L), suggesting enhanced hemicellulose breakdown. This behavior is consistent with previous studies, where acid pretreatment enhances hemicellulose solubilization, leading to increased release of pentose sugars such as xylose, while alkaline pretreatment reduces inhibitor formation (Świątek et al., 2020). During acid pretreatment, the hemicellulose is broken down, and its constituent sugars (mainly the pentoses xylose and arabinose) are released; therefore, these sugars are already present before enzymatic hydrolysis. This represents an advantage of acid pretreatment, although it becomes an important consideration when deciding whether to incorporate a washing step into the treatment. Low levels of cellobiose across treatments indicate efficient enzymatic conversion to glucose, whereas minor arabinose release (1.1-3.1 g/L) aligns with the known composition of grass hemicellulose.

The assessment of inhibitory compounds (Fig. 2-b) shows distinct variations among the pretreatment methods. Acid-pretreated hydrolysates contained elevated levels of HMF (1.3 g/L) and furfural (0.1 g/L), whereas acetic acid concentrations were highest in the unwashed base treatment (3 g/L), moderate in acid hydrolysates (2 g/L), and lowest in base-washed samples (0.8 g/L). Catechol was detected at low concentrations across all treatments, with slightly higher levels in the acid-pretreated samples. These findings are consistent with previous reports indicating that acid pretreatment facilitates the formation of furanic inhibitors, which can adversely affect downstream fermentation performance (Abedi and Hashemi, 2020; Yankov, 2022). Overall, acid pretreatment yielded the highest fermentable sugar release but generated substantial inhibitors, whereas base pretreatment avoided furan formation yet increased acetic acid levels, while the base-washed strategy provided a balanced compromise with moderately high sugar yields and minimal inhibitor accumulation. The limited change in sugar concentration after washing, despite a significant reduction in inhibitors, can be explained by the fact that washing primarily removes soluble inhibitory compounds, while the majority of fermentable sugars are generated during enzymatic hydrolysis from structural carbohydrates retained in the solid phase.

3.2. Lactic acid fermentation

The lactic acid fermentation performance of acid-, base-, and base-washed hydrolysates using *Heyndrickxia coagulans* demonstrated substantial differences in fermentation kinetics, lactic acid concentration, and overall productivity across the evaluated pretreatment strategies.

3.3. Final lactic acid concentrations

Fig. SI2 summarizes the final concentrations of lactic acid, acetic acid, and residual sugars following the fermentation of all hydrolysates. Among the test conditions, the acid-pretreated hydrolysate yielded the highest lactic acid concentration (36 g/L), medium acetic acid levels (2.6 g/L), and minimal residual sugars (0.6 g/L), demonstrating efficient sugar conversion despite the presence of acid-derived inhibitors. The obtained lactic acid concentrations are comparable to reported values for lignocellulosic hydrolysates fermented by *H. coagulans*, indicating competitive performance of grass biomass as a substrate (Ma et al., 2016; Susmozas et al., 2025). For example, lactic acid concentrations of up to 57 g/L have been reported from sugarcane bagasse using more complex process configurations such as pre-fermentation and simultaneous saccharification and co-fermentation (Qiu et al., 2023). In contrast, the base-pretreated hydrolysate produced a slightly lower lactic acid level of 31 g/L but retained the highest residual acetic acid (3.2 g/L), likely contributing to reduced fermentation efficiency. The base-washed hydrolysate generated 29.7 g/L lactic acid while exhibiting substantially lower acetic acid levels (1.3 g/L) and negligible residual sugars (0.7 g/L), indicating that the washing step effectively removed soluble inhibitors and improved microbial performance. When compared with the acetic acid concentration prior to fermentation (0.8 g/L), the slight increase observed indicates that fermentation had minimal influence on acetic acid levels. These results are consistent with recent studies highlighting the feasibility of producing lactic acid from lignocellulosic biomass as a sustainable alternative to conventional sugar-based processes (Díaz-Orozco et al., 2025).

In addition to lactic acid concentrations, fermentation efficiency can be further evaluated based on the yield of lactic acid relative to the available sugars. Using the initial sugar concentrations in the hydrolysates, the lactic acid yield was estimated to be approximately 0.80 g/g for the acid-pretreated sample, 0.84 g/g for the base-pretreated sample, and 0.80 g/g for the base-washed condition. These values approach the theoretical maximum yield for homofermentative lactic acid production (~1 g/g), indicating efficient conversion of sugars under all tested conditions (Di Lorenzo et al., 2022). Comparable studies have reported yields up to 90-97% of the theoretical maximum, confirming that such performance is achievable in lignocellulosic systems (Zhao et al., 2013). Notably, despite the presence of inhibitory compounds in the acid-pretreated hydrolysate, the yield remained comparable across all conditions, highlighting the robustness of the strain and the effectiveness of the fermentation process.

3.4. Fermentation kinetics

Fig. 3 illustrates the fermentation trend of the acid-pretreated hydrolysate, where lactic acid concentrations increased rapidly during the first 10 h before transitioning into a slower phase and eventually plateauing at 36 g/L by 26 h. This behavior is consistent with sequential sugar metabolism (diauxic growth), in which different monosaccharides are consumed successively, contributing to the multiple exponential

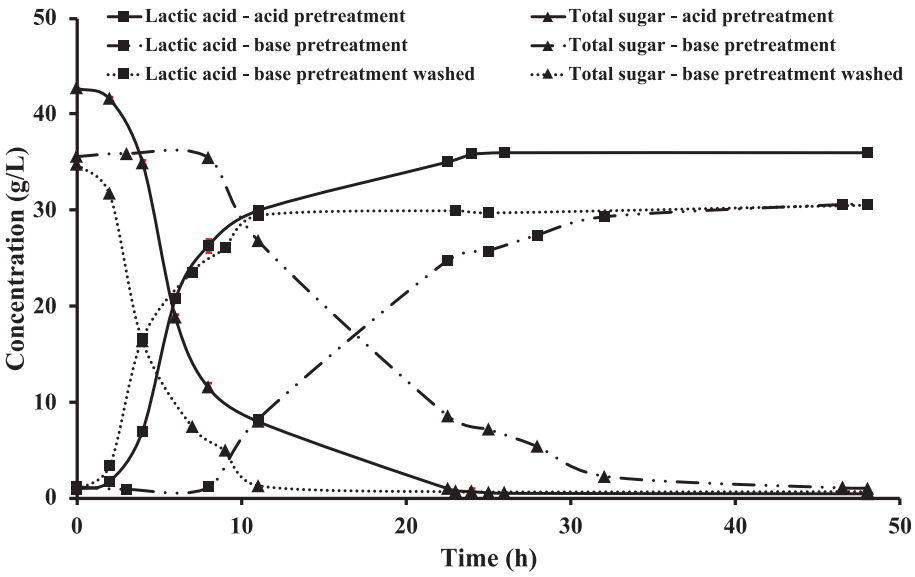


Fig. 3. Kinetics of lactic acid production and total sugar consumption (g/L) during fermentation of a) acid-, b) base-, and c) base-washed grass hydrolysates by H. coagulans.

phases observed. Simultaneously, total sugar levels declined from 42.7 g/L to nearly zero, indicating efficient substrate utilization. The extended time required to reach the plateau can be attributed to the presence of furanic inhibitors, such as HMF and furfural, generated during acid pretreatment, which slowed complete fermentation. Similar patterns have been reported for *Heyndrickxia coagulans* grown on acid-pretreated lignocellulosic hydrolysates, where furans prolong fermentation while still allowing high lactic acid titers, such as 45 g/L achieved by strain GKN316 (Jiang et al., 2016; Jönsson and Martín, 2016).

With the base-pretreated hydrolysate, lactic acid accumulation shows an evident lag (8–10 h) followed by a gradual rise to 30.6 g/L by 48 h. Sugar consumption closely followed this delayed trend, reflecting slower conversion kinetics. The prolonged fermentation period is likely attributed to acetate accumulation and lignin-derived soluble compounds generated during base pretreatment. Similar trends were reported in a previous study, where the fermentation process on alkaline-derived hydrolysates exhibited extended processing times despite achieving a relatively high final lactic acid concentration (Zhang et al., 2016).

In the base-washed hydrolysate, fermentation commenced immediately without a detectable lag phase, with nearly complete sugar depletion achieved within almost 11 h and lactic acid concentrations reaching 29.5 g/L. The absence of a delay in lactic acid production and the rapid consumption of sugars indicate that washing effectively removed soluble inhibitory compounds generated during pretreatment. This reduction in inhibitor load allowed *H. coagulans* to efficiently metabolize both hexose and pentose sugars, resulting in a faster and more streamlined fermentation process compared to the unwashed base-pretreated hydrolysate.

3.5. Volumetric productivity of lactic acid

Fig. SI3 illustrates the influence of pretreatment strategies on volumetric productivity during lactic acid fermentation. Among the tested conditions, the base-washed hydrolysate exhibited the highest productivity (2.6 g/L·h), representing a 1.6-fold improvement over the acid-pretreated hydrolysate (1.6 g/L·h) and a 2.9-fold increase compared to the unwashed base-pretreated hydrolysate (0.8 g/L·h). This enhanced performance highlights the effectiveness of the washing step in removing soluble inhibitors, enabling *H. coagulans* to utilize the available sugars more efficiently and achieve faster lactic acid accumulation. In addition, high lactic acid productivities have been reported in engineered systems, reaching up to 1.82 g/L·h under optimized conditions (Qiu et al., 2018). In comparison, the present study achieved a maximum productivity of 2.6 g/L·h without genetic modification or process intensification, highlighting the efficiency of the proposed approach under relatively simple process conditions. Further improvements in productivity may be achieved through process intensification strategies such as continuous fermentation, which has been reported as an effective approach to enhance volumetric productivity and is considered as a potential direction for future work (Schroedter et al., 2025). These results demonstrate that grass-derived hydrolysates can achieve competitive lactic acid production without complex process intensification, while preserving a chemically reactive solid fraction suitable for subsequent HTF upgrading.

3.6. Ions by-products after fermentation

Fermentation of grass-derived hydrolysates resulted in the release and accumulation of multiple inorganic ions originating from the raw biomass, chemical pretreatments, and cellular metabolism (**Fig. SI4**). Quantification of the major inorganic species revealed notable differences among the three pretreatment scenarios.

For acid pretreatment, relatively high concentrations of most ions were detected, including SO_4^{2-} (10,695 mg/L), K^+ (25,721 mg/L), Cl^- (728 mg/L), Na^+ (416 mg /L), and Ca^{2+} (512 mg/L). These elevated mineral loads likely originate from acid–mineral solubilization processes and release of mineral constituents from the lignocellulosic matrix. In comparison, base pretreatment resulted in moderately reduced ion concentrations. Sulfate and potassium decreased to 5,712 mg/L and 21,383 mg/L, respectively, while other ions remained within comparable magnitudes (e.g., Cl^- (795 mg/L), Na^+ (400 mg/L), Ca^{2+} 413 (mg/L)).

Substantial reductions in ionic contaminants were observed following base pretreatment with washing, where SO_4^{2-} decreased to 2,721 mg/L, K^+ to 15,593 mg/L, Na^+ to 269 mg/L, and Cl^- to 229 mg/L. This treatment, therefore, yielded the lowest overall ion load. Nitrogen species (NO_2 , NO_3 , NH_4^+) remained low across all conditions.

The presence of these ions is a significant consideration for process development. Elevated concentrations of monovalent and divalent ions increase the burden on downstream processing (DSP), which is necessary to obtain a food- or polymer-grade lactic acid product. Their

removal typically requires multistep purification strategies involving membrane separations, ion-exchange resins, evaporation, and crystallization operations (Pleissner et al., 2017). Each additional purification step increases operational complexity, energy consumption, and overall processing cost. Thus, minimizing inorganic impurities at earlier process stages can reduce downstream intensity and improve techno-economic performance. Considering the three pretreatment scenarios, the base-washed system presents the most favorable profile, with markedly reduced ionic by-products and consequently lower purification demand. In contrast, the acid-pretreated broth exhibits elevated sulfate and potassium levels, necessitating more intensive downstream purification.

Therefore, pretreatment strategies that inherently minimize inorganic release offer a tangible advantage by decreasing the scale and complexity of purification operations required to achieve high-purity lactic acid.

After separating the liquid hydrolysate for lactic acid fermentation, the remaining solid fraction, which, according to Table 1 still contains structural polysaccharides, lignin fragments, and minerals, was used as the feedstock for the next process step. Rather than discarding this residue, it was converted through hydrothermal fulvification, allowing further valorization of the biomass and improving overall carbon use. This integration enables the valorization of both liquid and solid process streams, thereby reducing waste generation and improving resource efficiency, while directing the residual solids toward the hydrothermal fulvification process described in the following section.

3.7. 15. Hydrothermal fulvification process

3.7.1. Elemental composition and yield of the solids and humic acids

The elemental analysis revealed a clear progression of chemical changes across the integrated process chain (Table 2). The untreated grass analyzed in this study contained approximately 46% carbon, with H/C and O/C ratios of 1.71 and 0.64, respectively, consistent with values reported for O-rich lignocellulosic biomass (Vassilev et al., 2010). After chemical pretreatment, both the acid-treated (AWS) and base-treated (BWS) residues showed a moderate decrease in C (to 41-42%) but significant O loss, reducing the O/C ratio to 0.48 while increasing H/C to 2.06. These changes indicate partial deoxygenation through hemicellulose solubilization and extractive removal. The substantial rise in ash content to more than 20%, is due to the loss of hemicellulose and extractives in the pretreatment (and thus a concentration of inorganics in the solid residue) and inorganic carryover from pretreatment chemicals. Such trends are consistent with observations in chemically pretreated agricultural residues, where oxygenated fractions are selectively removed, leaving more reduced and mineral-enriched solids (Awoyale et al., 2021).

HTF further intensified the transformations after the fermentation process, promoting dehydration and decarboxylation. The GF solid showed an increase in C to 51% and a decline in O/C to 0.26, reflecting strong deoxygenation under hydrothermal conditions. The decrease in O/C ratio reflects enhanced dehydration and condensation reactions characteristic of humification under alkaline hydrothermal conditions (Libra et al., 2011). Among the fermented residues, LAF displayed the most carbon-rich and condensed structure (C 56.1%, H/C 1.31, O/C 0.21), indicating efficient solid carbon retention and aromatic condensation, whereas LBF exhibited lower carbon content (35.2%) but high ash (46.8%), reflecting inorganic accumulation and dilution of the organic phase. The unwashed residues (LBF-NW-0.1 and 0.2) maintained intermediate compositions (C 43-44%, O/C 0.24-0.26) with increased yield at higher solid loading, suggesting enhanced carbon recovery rather than additional structural modification.

The recovered HA fractions, obtained at two different alkali concentrations of 0.1 and 0.2, achieve mass yields of 8.2 and 10.8 %, with a corresponding improvement in C, reaching 69.1 and 74.5 %. This observation indicates that alkaline severity promotes both enhanced carbon condensation and higher mass recovery of humic acids during HTF, a behavior in agreement with previously reported alkaline hydrothermal mechanisms, where extensive biomass fragmentation followed by secondary condensation drives the formation of HAs in the liquid phase (Ischia et al., 2026). Importantly, even after fermentation removes the carbohydrate fraction, the residual solids still produce A-HA yields comparable to, or slightly exceeding, those obtained from raw grass in standalone HTF (Ghobadian et al., 2026). This demonstrates that upstream lactic acid production does not diminish the humification potential of the biomass and confirms that the integrated fermentation-HTF approach enables additional value generation without compromising downstream humic substance recovery, while simultaneously reducing residual waste streams, in alignment with integrated biorefinery goals (Marzban et al., 2025). This is a key finding, as it demonstrates that upstream fermentation does not compromise the downstream humification potential, but instead enables a sequential valorisation pathway in which both major biomass fractions are effectively converted into value-added products.

The pronounced decrease in O/C from 0.16 to 0.09 at nearly constant H/C (1.42–1.45) indicates that oxygen removal is primarily governed by dehydration and condensation reactions rather than extensive aromatization. Under alkaline hydrothermal conditions, this reflects a reaction regime dominated by the fragmentation of biomass into reactive intermediates and their subsequent recombination. Carbohydrate-derived species undergo base-catalyzed isomerization followed by retro-aldol cleavage, generating low-molecular-weight carbonyl compounds and organic acids that define the primary reactive pool in the system (Esposito and Antonietti, 2013). With increasing alkalinity, the system shifts toward simplified, highly reactive intermediates, as base-tuned hydrothermal processing promotes the formation of small oxygenated molecules such as lactic, formic, and acetic acids while reducing overall molecular diversity (Tkachenko et al., 2023). In parallel, lignin undergoes extensive depolymerization through β-O-4 ether bond cleavage,

Table 2
Chemical composition of different feedstocks, hydrochars, and extracted A-HAs (C: carbon, H: hydrogen, N: nitrogen, S: sulfur, O: oxygen).

Sample	Ash (%)	C (%)	H (%)	N (%)	S (%)	O (%)	H/C	O/C	Yield (%)
Raw and pretreated materials									
Grass	7.1±0.3	45.9±1.2	6.5±0.3	1.1±0.1	0.4±0.1	39.0±1.5	1.71	0.64	-
AWS	23.1±0.8	41.2±1.0	7.1±0.4	1.4±0.1	0.9±0.1	26.3±1.2	2.07	0.48	-
BWS	21.5±0.7	41.7±1.1	7.2±0.3	1.7±0.2	0.5±0.1	27.5±1.3	2.06	0.49	-
HTF-derived solids									
GF	22.7±0.9	51.0±1.5	6.9±0.2	1.1±0.0	0.4±0.0	17.9±0.8	1.64	0.26	33.9±1.2
LBF	46.8±1.2	35.2±1.0	5.7±0.2	0.2±0.0	0.2±0.0	12.0±0.6	1.95	0.25	7.5±0.4
LAF	20.3±0.6	56.1±1.3	6.1±0.3	1.6±0.2	0.4±0.0	15.5±0.7	1.31	0.21	49.3±1.8
LBF-NW-0.1	34.3±0.9	43.8±1.2	5.4±0.2	0.9±0.1	0.4±0.0	15.2±0.6	1.48	0.26	21.2±0.9
LBF-NW-0.2	35.5±1.0	43.4±1.1	5.5±0.2	1.0±0.1	0.5±0.0	14.1±0.6	1.52	0.24	45.4±1.5
Extracted humic acids									
A-HA-0.1	5.6±0.2	69.2±1.6	8.3±0.2	2.1±0.0	0.3±0.1	14.5±0.7	1.45	0.16	8.2±0.3
A-HA-0.2	5.0±0.1	74.5±6.8	8.8±0.9	2.5±0.3	0.1±0.0	9.1±0.5	1.42	0.09	10.8±0.4

Yield refers to solid yield for hydrochar samples and humic acid yield (HAY) for A-HA samples.

releasing phenolic and aromatic fragments that coexist with carbohydrate-derived intermediates and contribute to the evolving reaction network (Ischia et al., 2026). These mixed intermediates subsequently recombine through base-catalyzed aldol condensation, dehydration, and cyclization reactions, leading to the formation of polycondensed humic-like macromolecules enriched in carboxylate and phenolic functionalities. As demonstrated in hydrothermal humification systems, this degradation–recondensation interplay produces oxygen-rich, partially aromatic yet still aliphatic structures that resemble natural humic substances (Ischia et al., 2026), indicating that artificial humic substance formation is governed by a dynamic balance between fragmentation and controlled recondensation rather than a simple concentration or precipitation process. Therefore, the formation of artificial humic substances from fermentation residues is not a simple concentration or precipitation phenomenon, but rather a dynamic transformation governed by alkaline hydrothermal chemistry, where fragmentation and controlled recondensation occur simultaneously; the implications of these transformations on the composition of process liquids are discussed in the following section.

3.8. Organic compounds distribution in process liquids

Fig. 4 shows the chemical composition of process liquids obtained from HTF under five conditions, highlighting distinct variations in carbon distribution, organic acid generation, sugar release, and aromatic compound accumulation as a function of pretreatment history and solid loading.

In Fig. 4-a, the highest total carbon (TC) and total organic carbon (TOC) contents were observed for LBF-NW-0.2 (TC 39 g/L and TOC 36.2 g/L), followed by LBF-NW-0.1 (TC 20.6 g/L and TOC 20.2 g/L), indicating that unwashed residues at higher solid loadings released more dissolved carbon into the liquid phase. After the extraction of artificial humic acids (A-HA), the TC and TOC values decreased to 14.5 and 14.3 g/L for LBF-NW-0.1, and to 22.9 and 22.4 g/L for LBF-NW-0.2, respectively, representing the carbon remaining in the artificial fulvic acid (A-FA) fraction. In contrast, GF (TC 20.5 g/L and TOC 19.6 g/L) and the washed residues (LBF and LAF) exhibited substantially lower TC and TOC levels, demonstrating that washing prior the fulvification process significantly reduced carbon solubilization potential. These findings are consistent with reports (Breulmann et al., 2018) that residual solubles and unhydrolyzed organics retained in unwashed solids enhance carbon recovery during HTF, especially at higher biomass concentrations.

***Fig. 4-b shows the distribution of organic acids, where a chromatographic peak assigned to lactic acid was detected in almost all process liquids, particularly in LBF-NW-0.1 (5.1 g/L) and LBF-NW-0.2 (9.9 g/L). Although residual lactic acid from upstream fermentation may

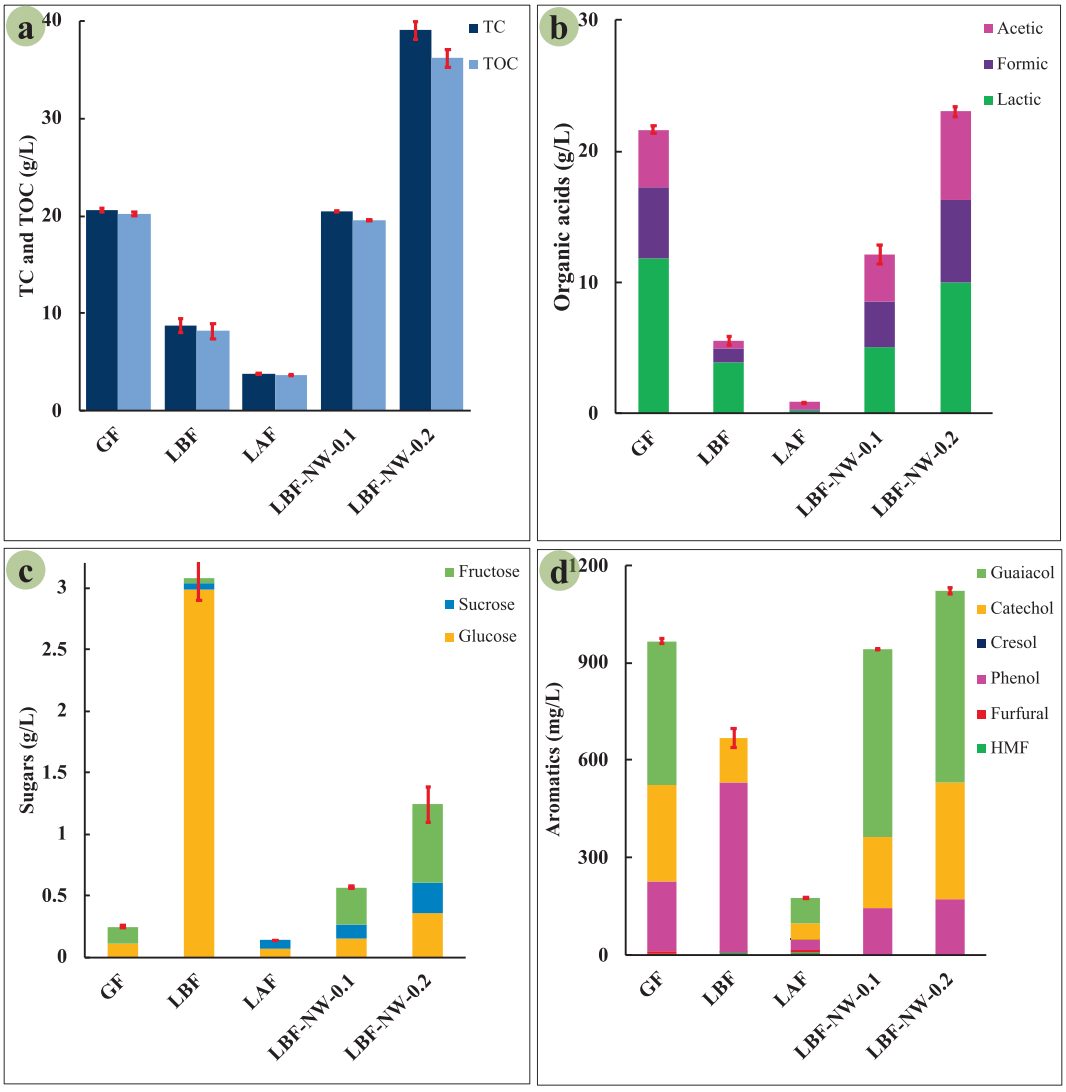


Fig. 4. The final concentration of a) TC and TOC, b)organic acids, c) sugars, and d) aromatics in the liquid products (g/L) from HTH process with different pretreatments.

contribute, its persistence alone is unlikely to fully explain the observed concentrations after HTF at 230 °C for 4 h. Acetic acid and formic acid concentrations were also highest in LBF-NW-0.2 sample (acetic acid 6.7 g/L and formic acid 6.3 g/L), reflecting a more extensive breakdown of hemicellulose and lignin derivatives at higher solid loadings. By comparison, LBF and LAF liquids exhibited lower organic acid levels, highlighting the effect of washing in removing pre-existing solubles.

Moreover, lactic acid is likely also produced in situ through reactions occurring during HTF. Indeed, under strongly alkaline conditions, carbohydrate-derived intermediates (e.g., glucose/fructose) undergo base-catalyzed isomerization followed by retro-aldol cleavage into C3 intermediates, which are subsequently converted to lactic acid through rearrangement and dehydration pathways (Esposito and Antonietti, 2013; Ischia et al., 2026). In addition, base-tuned hydrothermal processing has been shown to shift biomass conversion toward low-molecular-weight organic acids, including lactic acid, formic acid, and acetic acid, as part of the humification/fulvification pathway (Tkachenko et al., 2023). Previous studies have demonstrated that lactic acid (or lactate) can be formed under alkaline hydrothermal conditions at temperatures up to 300 °C (Yin et al., 2016). Therefore, its presence at 230 °C in the present study is mechanistically plausible under alkaline HTF conditions.

However, it is also likely that simultaneous formation and degradation reactions occur under these conditions, and thus the assignment of the chromatographic peak should be interpreted with caution. Future work will include complementary analytical techniques (e.g., LC–MS) to enable unambiguous compound identification and further validate this assignment. In addition, acetic and formic acids may originate from secondary degradation pathways, including the cleavage and oxidation of carbohydrate-derived intermediates and lignin fragments under alkaline conditions (Ischia et al., 2026; Tkachenko et al., 2023). The parallel increase in acetic and formic acids at higher solid loads (LBF-NW-0.2) is consistent with a greater extent of alkaline fragmentation reactions, affecting both carbohydrate and lignin-derived intermediates.

***Fig. 4-c illustrates that LBF (washed residue from base-pretreated hydrolysate) contains by far the highest soluble sugar concentration (3.08 g/L), dominated by glucose with only trace fructose and sucrose. All other liquids contain only minor sugar levels, indicating limited preservation of monomers under those conditions. The pattern suggests that washing base-pretreated residues prior to HTF left more hydrolyzable carbohydrate available or reduced catalytic carryover that would otherwise convert solubilized sugars to downstream products (e.g., organic acids or furans) during heating, whereas unwashed residues LBF-NW-0.1 (1.24 g/L) and LBF-NW-0.2 (0.57 g/L) and acid-derived solids yielded low residual sugars (0.14 g/L), consistent with more extensive secondary conversion during HTF. This behavior is likely driven by the enhanced accessibility of residual cellulose following alkaline pretreatment, which facilitates the hydrolysis of carbohydrates during HTF.

***Fig. 4-d showed that GF liquid had a high total aromatic concentration of 967.73 mg/L, dominated by guaiacol (442.9 mg/L), followed by catechol (297.4 mg/L) and phenol (215 mg/L). In contrast, using a base-pretreated grass hydrolysate (LBF) shifted the aromatic profile toward phenol, which increased to 522.3 mg/L and accounted for most of the total aromatics (667 mg/L), while guaiacol was largely absent. Acid pretreatment (LAF) resulted in a substantial suppression of aromatic formation, yielding the lowest total concentration (175.1 mg/L). Notably, omitting the washing step after fermentation (LBF-NW) markedly increased aromatic recovery, with the 0.1 solid ration case reaching 942 mg/L, closely resembling GF, and the 0.2 solid ratio case further increasing to 1121.3 mg/L. In both LBF-NW samples, guaiacol remained the dominant compound (579.7 and 590.9 mg/L for 0.1 and 0.2, respectively), accompanied by substantial catechol (216.26 and 356.9 mg/L) and phenol (146.1 and 173.6 mg/L).

Ghaslani et al. (2024) found that the type of pretreatment applied to the feedstock strongly influences aromatic compound formation during hydrothermal processing, reporting that pretreated biomass retained more residual lignin and soluble degradation products, which subsequently promoted higher phenolic release in the process liquids. Phenolic compounds are well-known products derived from lignin depolymerization. However, their formation cannot be attributed exclusively to lignin. Indeed, given the relatively low lignin content of the processed residues, carbohydrate-derived pathways likely contribute to their formation. Under strongly alkaline HTF conditions, residual glucose, cellulose, and hemicellulose fractions undergo hydrolysis and retro-aldol cleavage to aldehyde intermediates, which can then participate in aldol condensation and rearrangement reactions, yielding phenolic structures.

A similar mixed lignin-carbohydrate contribution to phenolic formation under alkaline hydrothermal conditions was recently observed in our previous work, where cellulose alone was measured to contribute directly to phenolic compound formation (Ischia et al., 2026).

Finally, the absence of furanic-derived compounds, such as 5-HMF, further distinguishes the HTF from classical hydrothermal carbonization, where acidic dehydration pathways favor furan formation.

3.9. Ions distribution in process liquids

The ionic composition of the HTF process liquids (Fig. 5) differed markedly among samples and was dominated by potassium (K) in all cases, reflecting the upstream alkali-based operations and the strong solubilization of K into the aqueous phase during HTF. The highest total ionic load occurred in LBF and GF, driven primarily by K at 12.04 g/L and 11.61 g/L, respectively, accompanied by measurable Na (0.09 g/L in LBF; 0.33 g/L in GF) and S (0.13 g/Land 0.22 g/L, respectively). In contrast, LAF contained substantially less K (1.15 g/L), consistent with its origin from washed fermentation residue associated with the acid route and therefore reduced alkali carryover into HTF. Instead, LAF was characterized by comparatively higher Ca (0.11 g/L) and a pronounced increase in P (0.03 g/L) relative to the other liquids (P from 0.0032 to 0.0035 g/L for GF, LBF, and the unwashed cases), suggesting enhanced mobilization of Ca and P components released from biomass during HTF under low-alkali conditions.

Across all samples, Mg remained minor ($\leq$ 0.05 g/L, highest in GF), and trace metals (Cu, Mn, Zn) were consistently low (generally 10^{-4} to 10^{-3} g/L), with Zn ranging from 0.00022 g/L (GF) to 0.00259 g/L (LBF), indicating that the major differences among liquids were governed

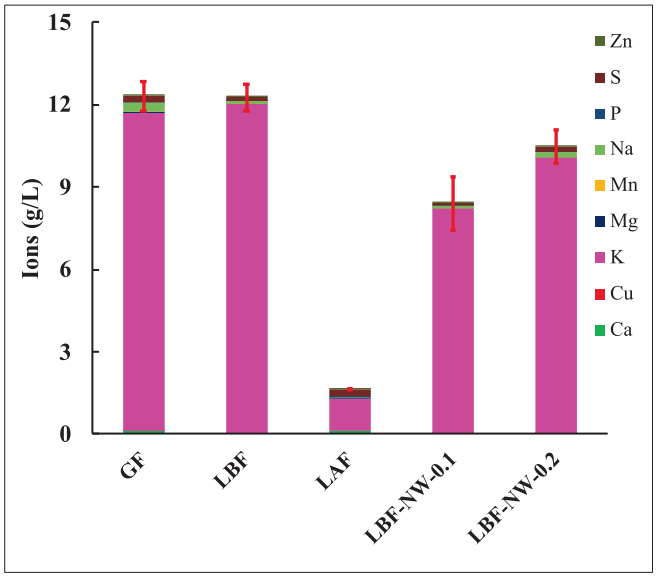


Fig. 5. The distribution of ions in the liquid products (g/L) from HTF process with different pretreatments.

primarily by alkali and counter-ions rather than trace metal dissolution.

The two unwashed residue cases reveal a clear loading-dependent carryover of soluble salts into HTF liquids. Increasing solids from LBF-NW-0.1 to LBF-NW-0.2 increased K from 8.2 to 10.1 g/L and Na from 0.10 to 0.21 g/L, alongside an increase in S from 0.08 to 0.17 g/L and a doubling of Cu from 0.000138 to 0.000278 g/L, while P remained essentially constant (0.0033 g/L) and Ca stayed low (0.0074-0.00785 g/L). This pattern aligns with the process history, as unwashed fermentation residues retain higher levels of water-soluble ions originating from upstream pretreatment, neutralization, and fermentation pH control, and higher solid loading increases the mass of retained solubles available for release during HTF.

By comparison, the washed residue liquids demonstrate that washing effectively reduced soluble salt content prior to HTF, relative to the K-dominated unwashed liquids (8.2-10.1 g/L), the acid-route washed residue produced a much lower K level (1.2 g/L), whereas the base-route washed residue still yielded very high K (12 g/L), consistent with stronger alkali association in the base-derived solids. Overall, the ionic trends across HTF process liquids closely reflect upstream treatment and washing strategies, with alkali-derived streams maintaining high K concentrations and unwashed, high-solids systems exhibiting enhanced salt release during hydrothermal processing.

The elevated potassium concentrations may be considered a limitation in conventional biorefinery schemes, particularly with respect to

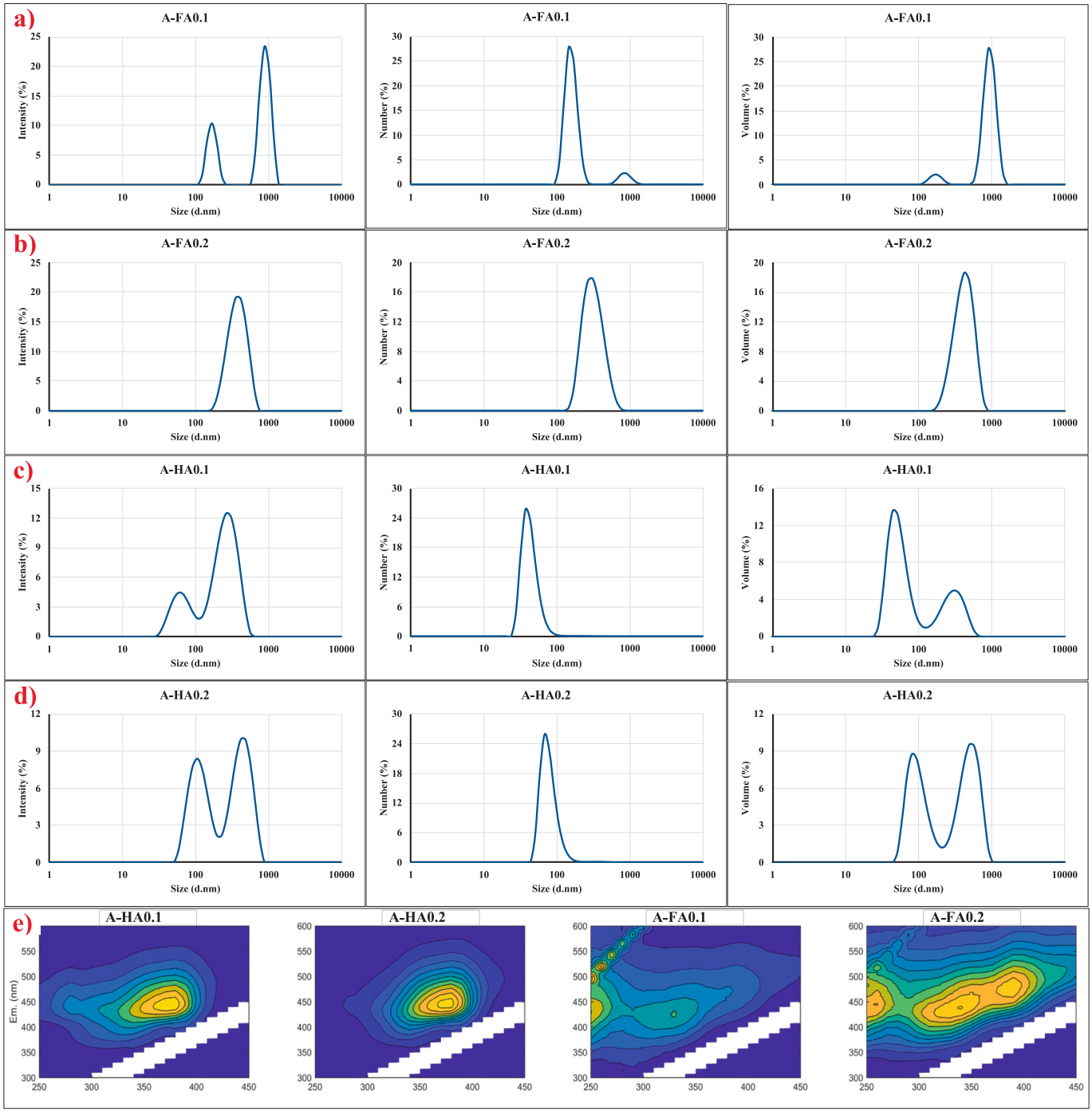


Fig. 6. Intensity-, number-, and volume-based particle size distributions of (a) A-FA0.1, (b) A-FA0.2, (c) A-HA0.1, (d) A-HA0.2, and (e) fluorescence EEM spectra showing humic-like signals in A-HA0.1 and A-HA0.2, and fulvic-like dominance in A-FA0.1 and A-FA0.2.

downstream purification of lactic acid, where high ionic strength increases the demand for separation processes such as ion exchange, membrane treatment, or crystallization. However, within the context of hydrothermal fulvification, this aspect can also be interpreted differently. Since the process is designed to convert the liquid phase into artificial humic substances rather than to purify individual compounds, extensive downstream salt removal is not necessarily required. Instead, the potassium-rich liquid phase contributes to the formation of potassium humate-like systems, which can be directly utilized after appropriate dilution.

In this context, the combined solid–liquid slurry obtained after HTF may be applied as a functional soil amendment, where both the organic humic components and the dissolved potassium provide agronomic value. This represents a key advantage of the HTF-based approach, as it reduces the need for energy- and cost-intensive separation steps typically required in conventional biorefineries. Similar concepts of utilizing humic-rich slurries without extensive purification have been reported in recent studies (Ghaslani et al., 2024; Sarlaki et al., 2026). Nevertheless, controlled application rates and soil-specific evaluation remain necessary to avoid excessive salinity or nutrient imbalance.

3.10. Characterization of A-HA and A-FA

After acidification of the solution (as described below Fig. 4-a), the resulting solid and liquid fractions were separated and labeled as artificial humic acids A-HA0.1 and A-HA0.2 for the solid fractions obtained from 0.1 and 0.2 solid loadings, respectively, and as artificial fulvic acids A-FA0.1 and A-FA0.2 for the corresponding liquid fractions. The particle size distribution analysis revealed distinct differences between A-HA and A-FA fractions following hydrothermal treatment of grass. In the liquid phase, the number-based distributions at both A-FA0.1 (Fig. 6-a) and A-FA0.2 (Fig. 6-b) solids were predominantly monomodal and centered on small particle sizes (100-200 nm), while the corresponding intensity- and volume-based distributions exhibited only minor contributions from larger aggregates. This indicates that the liquid fraction was dominated by small, soluble particles with only a limited presence of large structures.

In contrast, the solid fractions (Fig. 6-c for A-HA0.1, Fig. 6-d for A-HA0.2) displayed a more complex, bimodal character in their intensity- and volume-weighted distributions, with substantial populations of larger particles and aggregates ($\geq$400 nm) persisting across both solid loadings. The number-based distributions of the solids also shifted toward larger sizes relative to the liquids, confirming the prevalence of coarse structures in the solid phase. Since fulvic acids are known to occur as smaller, more soluble entities, whereas humic acids form larger, more aggregated structures, these findings strongly suggest that the liquid fractions are enriched in fulvic acids, while the solid residues are enriched in humic acids. This interpretation is consistent with the established molecular size differences between humic and fulvic acids reported in aquatic systems (Thurman et al., 1982).

The fluorescence EEM spectra (Fig. 6-e) provided clear evidence of compositional differences between A-HA and A-FA fractions obtained from hydrothermal fulvification. Samples A-HA0.1 and A-HA0.2 exhibited pronounced humic-like fluorescence (Peak C: Ex 330-370 nm, Em 420-480 nm), which is widely associated with high molecular weight, aromatic, and terrestrially derived humic substances (Chen et al., 2003). The A-HA0.2 sample showed a stronger and broader Peak C signal, consistent with greater aromatic condensation under higher solid loading. In contrast, A-FA0.1 and A-FA0.2 , displayed weaker humic-like features and additional protein-like signals (Peaks B/T), indicative of more labile, low molecular weight fluorophores such as fulvic acids (Coble, 1996).

The zeta potential measurements revealed pronounced differences between the liquid and solid fractions. The both samples A-FA0.1 ($+5.6 \pm 1.0$ mV) and A-FA0.2 ($+4.5 \pm 0.4$ mV) showed low, slightly positive surface charges, indicating weak electrostatic repulsion and relatively unstable colloidal suspensions. This is consistent with the predominance of small, fulvic-like molecules in the liquid phase, which are more soluble and less prone to forming stable, charged aggregates (Klučáková, 2018).

In contrast, the samples of A-HA0.1 (-41.3 ± 5.3 mV) and A-HA0.2 (-38.3 ± 10.1 mV) exhibited strongly negative surface charges, a signature of highly charged, stable colloidal particles. Such values are typical of humic acids, which possess abundant deprotonated carboxyl and phenolic groups that impart strong negative charges under neutral to alkaline conditions (Motta et al., 2016). The marked difference between the weakly positive liquid fractions and the strongly negative solid fractions, therefore, reinforces the earlier conclusion from particle size analysis: that the liquid phase is enriched in smaller, less aggregated fulvic acids, while the solid phase contains larger, more aromatic and aggregated humic acids.

The UV-Vis spectra of the A-HA and A-FA fractions (**Fig. SI5**) revealed distinct differences in their spectral profiles, consistent with the expected behavior of humic substances. The A-FAs showed higher absorbance in the UV region (200-220 nm) with a steep decline toward longer wavelengths, typical of less-condensed structures such as fulvic acids. In contrast, the A-HAs exhibited broader spectra with stronger absorption tails into the visible region (400-600 nm), characteristic of more aromatic and polymerized humic acid-like material. These qualitative spectral differences provide strong evidence for the enrichment of fulvic acids in the liquid fractions and humic acids in the solid residues.

The E4/E6 ratios (A465/A665) calculated from the spectra further support this interpretation, with values of 8.0 to 8.5 for the A-FAs and 2.7 to 4.0 for the A-HAs. Literature indicates that fulvic acids typically exhibit high E4/E6 ratios in the range of 12.0-15.5, reflecting their lower molecular weight and structural heterogeneity, while humic acids show lower ratios of 3.1-5.5, consistent with larger molecular size and higher aromatic condensation (Campanella et al., 1994; De Haan, 1993; Zou et al., 2021). Although the E4/E6 values of the A-FAs are slightly lower than those reported for standard fulvic acids, they clearly follow the same trend, falling between fulvic- and hymatomelanic acid values, while the A-HAs align closely with the humic acid range.

Collectively, these complementary techniques demonstrate that hydrothermal fulvification partitions the biomass-derived carbon into an aromatic, humic-like solid phase and a more labile, fulvic-like liquid phase, with the degree of humification increasing under higher total solids loading.

3.11. Waste-minimising approach and system-level considerations

The integrated fermentation–hydrothermal fulvification (HTF) system presented in this study is designed as a waste-minimising approach, in which all major biomass fractions are sequentially valorised rather than separated into discrete waste streams. Based on the experimental results, fermentation produced up to 36 g/L lactic acid, demonstrating efficient utilisation of the carbohydrate fraction. To achieve a waste-minimising system, the remaining fermentation solid residues must also be valorised, as they retain significant potential due to their lignin-rich composition. If left untreated, such semi-converted residues remain biologically unstable and prone to microbial degradation, potentially leading to greenhouse gas emissions (e.g., methane under anaerobic conditions) and carbon loss to the atmosphere (Song et al., 2023), as well as nutrient leaching and persistence of residual biological activity and potential pathogens (Maynaud et al., 2016). These residues are converted into artificial humic substances under alkaline hydrothermal fulvification conditions (Tkachenko et al., 2023). The elevated temperature and pressure during hydrothermal processing enable effective pathogen inactivation and degradation of microbially derived DNA, including antibiotic resistance genes (Ducey et al., 2017), an effect further enhanced by strongly alkaline conditions that promote degradation of cellular components and genetic material, resulting in a stable and sanitised product.

This approach transforms the solid residue into a functional product system consisting of hydrochar (humin-like fraction) and potassium humate-rich dissolved organic matter. The separation into solid and liquid fractions was performed solely for analytical purposes; in practical applications, the entire material may be utilised directly depending on the intended use. Under these conditions, HTF yielded 21–45 wt.% hydrochar and up to 10.8 wt.% humic acid, alongside a dissolved organic carbon-rich liquid phase (up to 36.2 g/L), demonstrating effective conversion of fermentation residues into value-added artificial humic substances. These results indicate substantial retention of biomass-derived carbon within solid and liquid phases, supporting cascading utilisation of both carbohydrate and lignin fractions. Although a complete mass and carbon balance was beyond the scope of this study, the reported yields and compositional data provide a consistent indication of carbon distribution across the main process outputs, in agreement with previous studies (Ghobadian et al., 2026; Marzban et al., 2024; Tkachenko et al., 2023).

To provide a clearer system-level interpretation in the absence of a full mass-balanced flow diagram, the main product streams can be summarised as follows: the carbohydrate fraction is primarily converted into lactic acid during fermentation (up to 36 g/L), while the remaining lignin-rich solids are transformed during HTF into hydrochar (21–45 wt. %) and artificial humic substances (up to 10.8 wt.%). The liquid phase contains dissolved organic carbon (up to 36.2 g/L), representing the fulvic-like fraction enriched in low-molecular-weight organics. These results describe the distribution of biomass-derived carbon across the integrated process and support a cascading valorisation pathway. This system-level interpretation highlights the advantage of the integrated approach compared to conventional single-product biorefineries, where lignin-rich residues are often underutilised or treated as waste streams.

Within this framework, the HTF liquid phase is not necessarily a residual waste stream but can function as an integral component of the product system. Unlike conventional biorefinery approaches that rely on downstream purification, this concept is based on artificial humic substances, where soluble and insoluble fractions collectively form a functional material for soil application (Ghaslani et al., 2024; Marzban et al., 2025). The potassium-rich liquid phase may be directly applied after appropriate dilution, providing nutrients while potentially avoiding additional separation or treatment steps (Ghaslani et al., 2024). Furthermore, the hydrothermal conditions promote the transformation or reduction of potentially harmful or inhibitory compounds (Marzban et al., 2024).

Notably, significant concentrations of lactic acid and other organic acids, including formic and acetic acids, are also present in the liquid phase (Stevenson, 1994; Tkachenko et al., 2023), and may contribute to the fulvic-like fraction, which comprises a heterogeneous mixture of low-molecular-weight organic compounds, including carboxylic acids and other oxygenated species (Stevenson, 1994). These organic acids should not be interpreted only as residual fermentation products, but also as characteristic intermediates and by-products of alkaline hydrothermal fulvification, where retro-aldol fragmentation and subsequent condensation reactions occur simultaneously. It is also worth noting that, under alkaline hydrothermal conditions, carbon losses to the gas phase may be significantly reduced with increasing alkalinity, and under HTF conditions gas formation can be strongly suppressed, with carbon largely retained in the solid and liquid product phases (Marzban et al., 2024). This contrasts with conventional hydrothermal carbonization (HTC), where decarboxylation reactions under non-alkaline conditions typically result in CO_2 formation (Funke and Ziegler, 2010; Libra et al., 2011). In this context, the solid, liquid, and humic fractions can be considered as part of a unified product system (Sarlaki et al., 2026); however, further research is required to evaluate the applicability of this approach across different feedstocks, process conditions, and end-use scenarios. Nevertheless, comprehensive system-level assessments, including full mass and carbon balances, as well as quantification of gas-phase carbon (e.g., CO_2), and validation under practical application conditions, are needed in future work. These aspects, along with techno-economic implications, are discussed in the following section.

3.12. Techno-economic considerations

From a techno-economic perspective, lactic acid production from lignocellulosic biomass is considered a competitive pathway, with recent studies reporting minimum selling prices in the range of approximately 993-1390 USD/ton depending on feedstock and process configuration (Manandhar and Shah, 2023). These costs are highly sensitive to fermentation yield, feedstock price, and process efficiency, with raw materials and utilities contributing up to 20-34% and 21-26% of total production costs, respectively (Manandhar and Shah, 2023). In addition, downstream separation and purification are major economic bottlenecks, accounting for up to 50% of total production cost in some processes (Gasca-González et al., 2019). In this context, the relatively high lactic acid yields obtained in the present study (0.8 g/g) represent a significant advantage, as yield has been identified as one of the most critical parameters influencing overall process economics. Furthermore, the use of grass as a low-cost and widely available feedstock may further improve economic competitiveness compared to sugar-based substrates.

The hydrothermal fulvification (HTF) step introduces additional operational costs associated with energy demand and chemical input. Hydrothermal conversion systems are known to be energy-intensive due to elevated temperature and pressure requirements, while alkaline processing further increases operational cost through alkali consumption. In techno-economic analyses of lignocellulosic systems, pretreatment and hydrothermal processing stages are consistently identified as major contributors to both capital and operating costs, largely due to reactor requirements and chemical usage (Manandhar and Shah, 2023). Therefore, the economic feasibility of the HTF step strongly depends on process optimization strategies such as heat integration, reduction of residence time, and recovery or recycling of alkaline streams. Without such measures, the combined energy and chemical demand may limit large-scale implementation.

Despite these challenges, the integrated design of the present system provides a distinct economic advantage by generating multiple value streams from a single feedstock. Conventional lactic acid processes typically leave lignin-rich residues underutilized or require additional waste treatment, whereas in this study these residues are converted into artificial humic substances. Previous techno-economic studies have shown that co-production of multiple products significantly improves overall profitability and reduces waste management costs in biorefinery systems (Kwan et al., 2018). In this context, the combined production of lactic acid and humic substances enhances carbon utilization and increases value recovery per unit biomass, which may offset the additional costs associated with HTF. However, further system-level assessment, including energy integration and chemical recovery efficiency, is required to fully quantify the economic potential of the proposed process.

4. Conclusions

This study demonstrated the multistage valorization of grass biomass through fermentation and hydrothermal fulvification (HTF), with pretreatment playing a decisive role in both stages. Acid-pretreated hydrolysates delivered the highest lactic acid titers (36 g/L) but required extended fermentation times due to furanic inhibitors, while base-pretreated hydrolysates showed slower conversion linked to acetate and lignin-derived solubles. In contrast, washed base-pretreated hydrolysates achieved rapid sugar utilization and lactic acid production (29.5 g/L) without a lag phase, confirming the benefit of removing inhibitory compounds.

The impact of pretreatment extended to HTF, where unwashed residues at higher solid loadings released the most dissolved carbon, organic acids, and aromatics, while washed residues minimized the

presence of solubilized inhibitors. Notably, the base-washed residue yielded the highest residual sugar concentrations, suggesting a more favorable balance between carbon preservation and inhibitor suppression. Collectively, the findings indicate that although acid pretreatment maximizes lactic acid yield, its downstream HTF performance is limited. In contrast, base-washed pretreatment offers a more integrated advantage by enabling efficient fermentation and cleaner HTF-derived liquids. A qualitative techno-economic assessment suggests that the integrated production of lactic acid and humic substances may improve overall process value, although detailed system-level evaluation is required. The combined strategy supports a waste-minimising biorefinery concept, demonstrating that fermentation and HTF can act as complementary processes to enable cascading biomass valorisation and improved overall carbon utilisation.

Author's contributions

Saman Ghobadian: Design of experiments, methodology, visualization, data collection, investigation, and writing the original draft of the manuscript. **Roland Schneider:** Design of experiments, methodology, review, and editing. **Giulia Ischia:** Interpretation, review, and editing. **Marcus Fischer:** Implementing of experiments, review, and editing. **Osvaldo Romero Romero:** Interpretation, review, and editing. **Joachim Venus:** Interpretation, review, and editing, **Thomas Hoffmann:** Interpretation, review, and editing. **Barbara Sturm:** Interpretation, review, and editing. **Vera Susanne Rotter:** Interpretation, review, and editing. **Matthias Kraume:** Supervision, review, and editing. **Nader Marzban:** Supervision, conceptualization, design of experiment, methodology, writing the original draft of the manuscript, review, and editing. All authors have read and approved the final version of the manuscript

CRediT authorship contribution statement

Saman Ghobadian: Writing – original draft, Visualization, Methodology, Investigation, Data curation. **Roland Schneider:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Giulia Ischia:** Writing – review & editing, Formal analysis. **Marcus Fischer:** Writing – review & editing, Investigation. **Osvaldo Romero Romero:** Writing – review & editing, Formal analysis. **Joachim Venus:** Writing – review & editing. **Thomas Hoffmann:** Writing – review & editing, Resources, Funding acquisition. **Barbara Sturm:** Writing – review & editing, Conceptualization. **Vera Susanne Rotter:** Writing – review & editing. **Matthias Kraume:** Writing – review & editing, Supervision, Conceptualization. **Nader Marzban:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.wasman.2026.115625.

Data availability

No data was used for the research described in the article.

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